

Analysis of photorefractive optical damage in lithium niobate: application to planar waveguides

J. Villarroel¹, J. Carnicero¹, F. Luedtke², M. Carrascosa¹, A. García-Cabañes^{1,*}, J. M. Cabrera¹, A. Alcazar³ and B. Ramiro³

¹Departamento de Física de Materiales C-IV, Universidad Autónoma de Madrid, E-28049, Spain

²Institute of Physics, University of Bonn, Wegelerstr. 8, 53115 Bonn, Germany

³Departamento Aerotecnia, EUIT Aerodúctica, Universidad Politécnica de Madrid, E-28040 Madrid, Spain

*angel.garcia@uam.es

Abstract: Photorefractive optical damage of single beams in LiNbO₃ crystals is analyzed within a framework of two photoactive centres (Fe²⁺/Fe³⁺ and Nb_{Li}⁴⁺/Nb_{Li}⁵⁺). It compares model simulations and significant experimental measurements in LiNbO₃ waveguides. A good agreement is found in the performed comparisons: photovoltaic currents, refractive index changes and, especially relevant, in degraded beam-profiles. The progress of the degraded wavefront has been simulated by implementing a finite-difference beam-propagating method which includes the model equations. These results, together with previous ones on grating recording, provide a comprehensive, satisfactory explanation of most important questions on photorefractive optical damage.

©2010 Optical Society of America

OCIS codes: (160.3730) Lithium niobate; (160.4330) Nonlinear optical materials; (160.5320) Photorefractive materials;

References and links

1. T. Volk, M. Wöhlecke, and N. Rubinina, "Optical damage resistance in lithium niobate", in *Photorefractive materials and their applications 2, Materials*, P. Günter and J. P. Huignard, eds, (Springer, 2007), pp165–203.
2. D. Kip, and M. Wesner, "Photorefractive waveguides", in *Photorefractive materials and their applications 1, Basic effects*, P. Günter and J.P. Huignard, eds, (Springer, 2006), pp 281–315.
3. D. S. Hum, and M. M. Fejer, "Quasi-phases matching," *C. R. Phys.* **8**(2), 180–198 (2007).
4. M. Kösters, B. Sturman, P. Werheit, D. Haertle, and K. Buse, "Optical cleaning of congruent lithium niobate crystals," *Nat. Photonics* **3**(9), 510–513 (2009).
5. A. Ashkin, G. D. Boyd, J. M. Dziedzic, R. G. Smith, A. A. Ballman, and K. Nassau, "Optically-induced refractive index inhomogeneities in LiNbO₃ and LiTaO₃," *Appl. Phys. Lett.* **9**(1), 72–74 (1966).
6. P. Gunter, and J. P. Huignard, eds., *Photorefractive Materials and Their Applications* (3 Vols.) (Springer Series in Optical Sciences, New York, 2006).
7. T. Volk, N. Rubinina, and M. Wohlecke, "Optical-damage-resistant impurities in lithium-niobate," *J. Opt. Soc. Am. B* **11**(9), 1681–1687 (1994).
8. O. Eknayan, H. F. Taylor, W. Matous, and T. Ottinger, "Comparison of photorefractive damage effects in LiNbO₃, LiTaO₃ and Ba_{1-x}Sr_xTi_yNb_{2-y}O₆ optical waveguides at 488 nm wavelength," *Appl. Phys. Lett.* **71**, 3051–3053 (1997).
9. D. A. Bryan, R. Gerson, and H. E. Tomaschke, "Increased optical damage resistance in lithium niobate," *Appl. Phys. Lett.* **44**(9), 847–849 (1984).
10. W. M. Young, R. S. Feigelson, M. M. Fejer, M. J. F. Digonnet, and H. J. Shaw, "Photorefractive-damage-resistant Zn-diffused waveguides in MgO:LiNbO₃," *Opt. Lett.* **16**(13), 995–997 (1991).
11. M. Asobe, O. Tadanaga, T. Yanagawa, H. Itoh, and H. Suzuki, "Reducing photorefractive effect in periodically poled ZnO- and MgO-doped LiNbO₃ wavelength converters," *Appl. Phys. Lett.* **78**(21), 3163–3165 (2001).
12. J. Rams, A. Alcázar de Velasco, M. Carrascosa, J. M. Cabrera, and F. Agulló-López, "Optical damage inhibition and thresholding effects in lithium niobate above room temperature," *Opt. Commun.* **178**(1-3), 211–216 (2000).
13. M. Falk, T. Woike, and K. Buse, "Reduction of optical damage in lithium niobate crystals by thermo-electric oxidation," *Appl. Phys. Lett.* **90**(25), 847–849 (2007).
14. J. Carnicero, M. Carrascosa, A. Méndez, A. García-Cabañes, and J. M. Cabrera, "Optical damage control via the Fe²⁺/Fe³⁺ ratio in proton-exchanged LiNbO₃ waveguides," *Opt. Lett.* **32**(16), 2294–2296 (2007).

15. J. C. Chon, W. Feng, and A. R. Mickelson, "Photorefractive damage threshold in Ti-LiNbO₃ channel waveguides," *Appl. Opt.* **32**(36), 7572–7580 (1993).
16. A. V. Ilyenkov, A. I. Khizniak, L. V. Kreminskaya, M. S. Soskin, and M. V. Vasnetsov, "Birth an evolution of wave-front dislocations in a laser beam passed through a photorefractive LiNbO₃:Fe crystal," *Appl. Phys. B* **62**(5), 465–471 (1996).
17. E. Jermann, and J. Otten, "Light-induced charge transport in LiNbO₃:Fe at high light intensities," *J. Opt. Soc. Am. B* **10**(11), 2085–2092 (1993).
18. M. Carrascosa, J. Villarroel, J. Carnicero, A. García-Cabañes, and J. M. Cabrera, "Understanding light intensity thresholds for catastrophic optical damage in LiNbO₃," *Opt. Express* **16**(1), 115–120 (2008).
19. F. Devaux, J. Safioui, M. Chauvet, and R. Passier, "Two-photoactive-center model applied to photorefractive self-focusing in biased LiNbO₃," *Phys. Rev.* **81**(1), 013825 (2010).
20. J. Carnicero, O. Caballero, M. Carrascosa, and J. M. Cabrera, "Superlinear photovoltaic currents in LiNbO₃ analyses under the two-center model," *Appl. Phys. B* **79**(3), 351–358 (2004).
21. N. Iyi, K. Kitamura, F. Izumi, K. Yamamoto, T. Hayasi, H. Asano, and S. Kimura, "Comparative study of defects structures in lithium niobate with different compositions," *J. Solid State Chem.* **101**(2), 340–352 (1992).
22. N. Zotov, H. Boysen, F. Frey, T. Metzger, and E. Born, "Cation substitution models of congruent LiNbO₃ investigated by X-ray and neutron powder diffraction," *J. Phys. Chem. Solids* **55**(2), 145–152 (1994).
23. M. Simon, St. Wevering, K. Buse, and E. Krätzig, "The bulk photovoltaic effect of photorefractive LiNbO₃:Fe crystals at high light intensities," *J. Phys. D* **30**(1), 144–149 (1997).
24. P. Herth, D. Schaniel, Th. Woike, T. Granzow, M. Imlau, and E. Krätzig, "Polarons generated by laser pulses in doped LiNbO₃," *Phys. Rev.* **71**(12), 125128 (2005).
25. G. de la Paliza, O. Caballero, A. García-Cabañes, M. Carrascosa, and J. M. Cabrera, "Superlinear photovoltaic currents in proton exchanged LiNbO₃ waveguides," *Appl. Phys. B* **76**, 555–559 (2003).
26. O. Caballero-Calero, J. Carnicero, A. Alcazar, G. de la Paliza, A. García-Cabañes, M. Carrascosa, and J. M. Cabrera, "Light-intensity measurements in optical waveguides using prism couplers," *J. Appl. Phys.* **102**(7), 074509 (2007).
27. F. Luedtke, J. Villarroel, A. García-Cabañes, K. Buse, and M. Carrascosa, "Correlation between photorefractive index changes and optical damage thresholds in z-cut proton-exchanged-LiNbO₃ waveguides," *Opt. Express* **17**(2), 658–665 (2009).
28. J. Villarroel, M. Carrascosa, A. García-Cabañes, and J. M. Cabrera, "Light intensity dependence of holographic response and dark decays in α -phase PE:LiNbO₃ waveguides," *J. Opt. A, Pure Appl. Opt.* **10**(10), 104008 (2008).
29. D. Berben, K. Buse, S. Wevering, P. Herth, M. Imlau, and T. Woike, "Lifetime of small polarons in iron-doped lithium-niobate crystals," *J. Appl. Phys.* **87**(3), 1034–1041 (2000).
30. O. F. Schirmer, M. Imlau, C. Merschjann, and B. Schoke, "Electron small polarons and bipolarons in LiNbO₃," *J. Phys. Condens. Matter* **21**(12), 123201 (2009).
31. G. Lifante, *Integrated Photonics: Fundamentals* (Wiley, Wiltshire, 2003).
32. J. Ramiro-Diaz, and A. A. de Velasco, "Orthogonal mode decomposition for efficient BPM applied to optical waveguides," *Ferroelectrics* **390**(1), 71–78 (2009).
33. F. Luedtke, *Optical damage in proton-exchanged waveguides in lithium niobate crystals*, Diploma Thesis, Bonn University (2008).
34. R. Göring, Y. L. Zhan, and St. Steinberg, "Photoconductivity and photovoltaic behaviour of LiNbO₃ and LiNbO₃ waveguides at high optical intensities," *Appl. Phys., A Mater. Sci. Process.* **55**, 97–100 (1992).

1. Introduction

The demand of photonic devices for medium and high optical power is continuously rising, both in scientific and technological fields. This implies strict design requirements to improve the device resistance to optical damage [1]. In the important group of electrooptic materials, the optical damage is mostly due to its associated photorefractive sensitivity that gives rise to a wavefront distortion of the laser-beam propagating through the material. In the case of LiNbO₃ (the reference material in electrooptics and integrated optics), the beam divergence associated to the wavefront distortion is a serious drawback for high light intensity applications [2–4]. On the discovery of the photorefractive effect in LiNbO₃, early in 1966 [5], it was realized that the crystal could be reverted to its initial condition by homogeneous illumination, indicating that a reversible refractive index change had taken place. Since then, a huge amount of papers have been published on both its physical features and exciting applications to photonic devices (see [6] and references therein). However, the beam-distortion aspect of photorefractive (optical damage) continues to be a crucial limiting factor in high and medium power photonic devices in LiNbO₃ and other photorefractive materials [7,8].

There have been partial advances in the practical control and reduction of photorefractive beam-distortion. First, the Fe concentration in the crystals is reduced as much as possible (down to a few atoms per million) because of its proved exceptional photorefractive efficiency. Further damage reductions have been achieved by some metal doping (specially Mg, Zn, In) [7,9–11], by increasing the operating temperature [12], by very strong crystal oxidation (i.e. negligible $[\text{Fe}^{2+}]/[\text{Fe}^{3+}]$ ratio) [13] or sample reduction [14] and by optical cleaning [4]. Nonetheless, with most of these methods it is difficult to predict the material behavior after the procedure followed in the fabrication of a particular device (the resulting optical damage often is unpredictable). From the theoretical side, there have been very few attempts in the past [15,16] addressed to describe photorefractive beam-distortion in LiNbO_3 , in spite of its scientific and technological relevance. The rough explanation of the beam distortion is very simple: the light induces a refractive index decrease Δn (via charge transport and electrooptic activity in the crystal) which gives rise to light self-diffraction or self-defocusing. However, a more detailed analysis shows the complexity of the subject as it involves the microscopic origin of the charge transport and its interaction with the wavefront change via the index change Δn .

Here we present a comprehensive analysis of significant experiments and simulations related to the photorefractive beam distortion. Namely, photovoltaic currents, light-induced changes of refractive index and wavefront evolution as the beam-intensity grows. A two-centre model with the defects $\text{Fe}^{2+}/\text{Fe}^{3+}$ and $\text{Nb}_{\text{Li}}^{4+}/\text{Nb}_{\text{Li}}^{5+}$, which has formerly been successful on dealing with grating recording [17,18], is implemented. The physics underlying the phenomenon is greatly clarified and all experimental results are satisfactorily explained. It also provides an efficient, reliable tool to improve the optical damage resistance of real photonic devices. Although the work is focused on LiNbO_3 waveguides for reasons given below, the general principles can be extended to bulk LiNbO_3 and likely to other photorefractive media. In fact, a recent paper [19] makes use of the same scheme to study the behavior of photorefractive solitons.

2. Theoretical model

In a similar manner as performed in [18], we assume that the secondary defect $\text{Nb}_{\text{Li}}^{4+}/\text{Nb}_{\text{Li}}^{5+}$ plays a central role in the process. This modifies the predictions based on the conventional model of a unique $(\text{Fe}^{2+}/\text{Fe}^{3+})$ -defect and becomes particularly relevant at high intensities. The main features of the secondary center, Nb_{Li} , as inferred from the superlinearity of the photovoltaic current [20], are the following. A high concentration (1%–3%) [21,22], a photovoltaic constant (upon $\text{Nb}_{\text{Li}}^{4+}$ excitation) about a factor 4.5 greater than that of the Fe^{2+} -defect [23], and a low activation energy for thermal electron detrapping ~ 0.4 eV [24]. Thus, at room temperature and in dark, practically all Nb_{Li} are in the $\text{Nb}_{\text{Li}}^{5+}$ -state.

The rate and transport equations for the two center model read [18]:

$$\frac{\partial n}{\partial t} = (S_1 I + S_{r1})N_1 + (S_2 I + S_{r2})N_2 - S_r n(N_{D1} - N_1 + N_{D2} - N_2), \quad (1a)$$

$$-\frac{\partial N_1}{\partial t} = \frac{\partial(N_{D1} - N_1)}{\partial t} = (S_1 I + S_{r1})N_1 - S_r n(N_{D1} - N_1), \quad (1b)$$

$$-\frac{\partial N_2}{\partial t} = \frac{\partial(N_{D2} - N_2)}{\partial t} = (S_2 I + S_{r2})N_2 - S_r n(N_{D2} - N_2), \quad (1c)$$

$$\vec{j} = eI(L_1 S_1 N_1 + L_2 S_2 N_2)\vec{u}_c + e\mu n\vec{E} - eD\nabla n, \quad (1d)$$

$$\nabla \cdot \vec{E} = \frac{e}{\epsilon\epsilon_0}(N_{D1} - N_1 + N_{D2} - N_2). \quad (1e)$$

Here, subscripts 1,2 stand for primary and secondary centers, n , $N_{1,2}$ and $N_{D1,D2}$ for the electronic carrier, donor and total center concentrations respectively and S_r is the electron recombination coefficient, which, for the sake of simplicity, is assumed to be the same for both types of traps. $S_{1,2}$ and $L_{1,2}$ denote the photo-ionization cross sections and the effective photovoltaic lengths respectively. $S_{ii} = S_{ii0} \exp(-\varepsilon_{ii}/KT)$ ($i = 1,2$) is the donor thermal ionization probability, and $\mu = eD/KT$ is the electron mobility with $D = D_0 \exp(-\varepsilon_D/KT)$ the diffusion coefficient. Finally, E is the space-charge electric field that generates the refractive index change through the electrooptic effect and u_c represent a unit vector along the polar c-axis. Note that, to simplify, direct electron transfer from Fe^{2+} to $\text{Nb}_{\text{Li}}^{5+}$ proposed in Ref [17], has been disregarded, taking into account the low Fe-concentration of nominally pure LiNbO_3 .

Table 1: Material parameters used for the simulations.

Parameter	Value	Ref.
N_{D1} (m^{-3})	2×10^{23}	This paper
N_1 (m^{-3})	$0.065 \times N_{D1}$	This paper
N_{D2} (m^{-3})	2×10^{26}	[21,22]
S_{ii0} (s^{-1})	5×10^{14}	[18]
ε_1 (eV)	1	[18]
S_{20} (s^{-1})	5×10^{11}	[18]
ε_2 (eV)	0.32	[20], [25], this paper
s_1 (m^2)	2×10^{-22}	[23]
s_2 (m^2)	1.5×10^{-22}	[29], [30]
L_1 (m)	5×10^{-10}	[23]
L_2 (m)	30×10^{-10}	[20], [25], this paper
s_r	1×10^{-15}	[18]
D_0 (m^2s^{-1})	1.6×10^{-5}	[18]
ε_D (eV)	0.16	[18]

In order to compare simulated and experimental results, the knowledge of the material parameters appearing in Eqs. (1a)-(1e) is essential. On dealing with grating recording, the parameter set choice in [18] resulted in a quite acceptable semi quantitative agreement between theory and experiments. For the single beam configuration of the present paper, we have-refined some parameter values according to new, more accurate experimental data. In particular, the measurements of photovoltaic currents reported in [20,25] have been here repeated taking into account the method of Ref [26]. for determining the beam intensity. From these data a more accurate value for the product $L_2 S_2$ has been obtained resulting equal to $4.5 \times L_1 S_1$, a value very similar to that previously reported for bulk LiNbO_3 [23]. From the temperature dependence of the photovoltaic current the refined activation energy for thermal detrapping of $\text{Nb}_{\text{Li}}^{4+}$ in waveguides now is $\varepsilon_2 = 0.32$ eV (instead of 0.4 eV) and the corresponding pre-exponential factor has been readjusted. It is worth emphasizing that the last, strongest support for the parameter choice presented in Table 1 is the good accordance between simulations and measurements observed in a wide variety of photorefractive experiments. For waveguides, they include holographic and single beam measurements in terms of quantities such as photovoltaic current [20,25], refractive index change [27], grating recording/decay [28] and high intensity thresholds for optical damage [14]. In addition, simulations and measurements include variables such as beam intensity (from 10^{-1} W/cm² to 10^3 W/cm²), temporal evolution, temperature (from 20°C to 150°C) and oxidation/reduction

state ($[\text{Fe}^{2+}]/[\text{Fe}^{3+}]$ from 0.05 to 2). Note that simulations also require to know N_{D1} ($[\text{Fe}]$) and N_1 ($[\text{Fe}^{2+}]$) values which depend on the sample origin. The values shown in Table 1 correspond to our proton exchange waveguides.

Using Eqs. (1a)-(1e), all relevant quantities (photovoltaic current j_{pv} , light-induced refractive index change Δn , photo- and electronic-conductivity σ , ...) can be numerically computed in terms of the light intensity and other significant variables (temperature, doping, reduction state, ...). In order to make the analysis of major practical interest for device designers, we have implemented a finite-difference Beam Propagation Method (BPM) [31] appropriate for the above model. It permits a detailed description of the wavefront distortion during propagation.

For all experiments and simulations carried out, the following planar-guide configuration has been used. Light confinement is along the lattice Z-axis (z -cut sample) with TM-polarization, beam propagation is along Y-axis while beam spreading takes place along X-axis. For the model application, the planar-guide configuration has the following advantages. It only needs a two-dimensional treatment, since the wavefront only spreads along the guide plane, and recently reported experimental data on optical damage in planar guides [20,25,27,28] can be used for comparison. The experimental advantages are that reaching high intensities is easier inside the guide because of the light confinement, and that this configuration is used in integrated optic devices. Furthermore, holographic induced nonlinear scattering can be neglected with z -cut guides, because recording of noise gratings with K-vectors parallel to the c -axis is strongly suppressed in the small guide thickness.

3. Results

Here we present three main results about dependences on the beam intensity. a) The saturating refractive index change (Δn_{sat}), which is directly responsible for the beam distortion. b) The bulk photovoltaic current (j_{pv}) and the electronic conductivity (σ), responsible for the charge transport..c) The beam-distortion itself, of major interest in practical devices.

3.1. Light intensity dependences of important quantities (j_{pv} , σ , Δn_{sat})

In order to calculate the total current density with Eq. (1d), its last term has been neglected since the derivative along X-axis of the beam intensity profile is quite small and the light intensity along the Z -axis is assumed to be constant. Then, the photovoltaic current density and so, the total current density, are along the Z- axis (perpendicular to the guide plane) and Eq. (1d) can be written:

$$j = eI(L_1 S_1 N_1 + L_2 S_2 N_2) + e\mu n E \quad (2)$$

Note that vector notation is no more necessary and has been removed. In turn, the space-charge field responsible for the saturating index change, Δn_{sat} , in this configuration is essentially coincident with the photovoltaic field (perpendicular to the guide plane) associated to the local intensity. At saturation, the first term of (2) (photovoltaic contribution) and the second one (space-charge field developed on illumination) equalize and the saturating space charge field writes:

$$E_{\text{sat}} = \frac{e(L_1 S_1 N_1 + L_2 S_2 N_2)I}{e\mu n} = \frac{j_{\text{pv}}}{\sigma} \quad (3)$$

The conductivity $\sigma = e\mu n$ includes the photo-induced and the dark conductivities. The dependences of N_1 , N_2 and n on the light intensity are obtained by numerical integration of Eqs. (1a)-(1c) [20]. From (3) Δn_{sat} writes:

$$\Delta n_{sat} = -\frac{1}{2}n^3 r_{33} E_{sat} \quad (4)$$

where r_{33} is the appropriate Pockels coefficient. Note that, as it is well known, the photorefractive index change for LiNbO₃ is negative.

The simulated logarithmic plot of $|\Delta n_{sat}|$ as a function of the light intensity has been drawn in Fig. 1 in continuous line. At very low intensities $I < 10^{-2}$ W/cm², a sloping-region is observed due to the competition between dark conductivity and the photovoltaic current. Here, $|\Delta n_{sat}|$ is too small to induce any measurable beam-distortion. Then, up to $I \approx 10$ W/cm², the photovoltaic current clearly dominates over dark conductivity and $|\Delta n_{sat}|$ is intensity independent as predicted by the conventional one-centre model. From $I > 10$ W/cm² (vertical dashed line in Fig. 1) $|\Delta n_{sat}|$ strongly increases with I , because the contribution of the secondary center to the photovoltaic current is quickly growing. Experimental values of $|\Delta n_{sat}|$ vs. the beam intensity I inside the guide, measured in [27] and [33] with a Mach-Zehnder interferometer technique for two similar proton-exchanged LiNbO₃ waveguides, have been represented as solid circles. The greatest intensity which could reliably be measured ($\sim 6 \times 10^2$ W/cm² for the sample length used) is limited by the beam spread. Up to this value, a good agreement is observed between experiments and simulations.

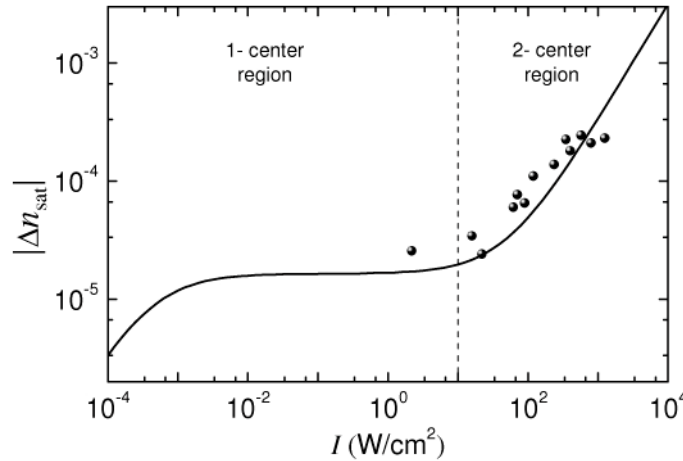


Fig. 1. Logarithmic plot of the calculated saturating index change $|\Delta n_{sat}|$ versus the beam intensity inside the waveguide (continuous line). The vertical dashed line indicates the onset of the two-centre regime. Experimental data (solid circles) have been measured in proton-exchanged LiNbO₃ waveguides (redrawn from [27] and [33]). For $I > 6 \times 10^2$ W/cm², the photorefractive beam-distortion makes unreliable any intensity measurement. The theoretical curve has been calculated with material parameters of Table 1.

In order to better understand the origin of the general features of Fig. 1, the intensity dependences of the photovoltaic current, and the total conductivity, σ , are now discussed. The calculated dependences with the same material parameters of Fig. 1, are shown in Fig. 2 as logarithmic plots. For comparison, corresponding experimental data of $j_{pv}(I)$ measured for α -phase guides with the same set-up as in [20], are represented in the inset of Fig. 2 together with the theoretical curve. Linear scales have been used here to better appreciate the superlinearity. The lowest experimental photocurrent is determined by the electrometer sensitivity, while the highest beam intensity is limited by the appearance of beam-distortion which leads to unreliable intensity measurements. j_{pv} exhibits a superlinear dependence on I within the range $10 \text{ W/cm}^2 < I < 2 \times 10^3 \text{ W/cm}^2$. This is a transition region between the low intensity regime, where the one-center (Fe) model is approximately valid, and the high intensity regime, where the contribution of the secondary center (Nb_{Li}) is dominant and the two-center model is required. In turn, σ presents a flat initial region for $I < 10^{-4}$ W/cm² where

dark-conductivity dominates over photo-conductivity, a linear region where the one-center model is valid, and a sublinear dependence within the range $2 \times 10^2 \text{ W/cm}^2 < I < 10^5 \text{ W/cm}^2$ where the two-center model is needed. Both nonlinear dependences in Fig. 2 arise from the growing contribution of the secondary center to j_{pv} and σ with increasing intensity and both nonlinearities contribute to the strong growth of $|\Delta n_{\text{sat}}|$, although the start of the increase only overlaps with that of j_{pv} .

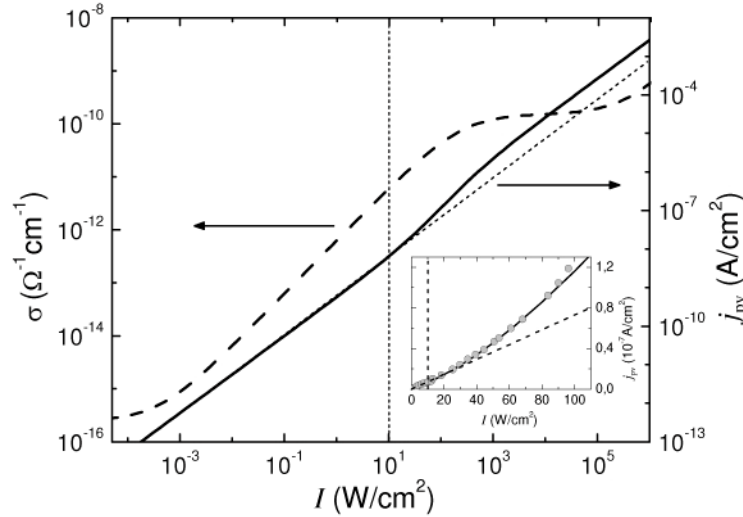


Fig. 2. Logarithmic plot of the simulated (computed) electronic conductivity σ (dashed line, left axis) and photovoltaic current j_{pv} (solid line, right axis) as a function of the beam intensity inside the waveguide with the same parameters of Fig. 1. For comparison, measured experimental data for $j_{pv}(I)$ are represented as solid circles in the inset together with the theoretical curve. To better appreciate the superlinearity we use here linear axis. In both, the figure and the inset, the dotted line represents the extrapolation of the linear dependence appearing at low I and the vertical line indicates the transition region from the one-center to the two-center regime.

A very good agreement is observed in Fig. 2 between calculated and experimental superlinear behaviors of j_{pv} in the proton-exchanged waveguide. A similar behavior has been observed in bulk LiNbO_3 [23]. Reported data on the experimental σ sublinearity [34] cannot be plotted for comparison because of the lack of sufficient parameter information, although the general trends look similar. Nonetheless, the dark conductivity value (flat region for $I < 10^{-4} \text{ W/cm}^2$ with $\sigma \sim 2 \times 10^{-12} \text{ } \Omega^{-1} \text{ m}^{-1}$) coincides within the experimental error with the reported experimental data [27]. Finally, the relevance of the two-center framework is further emphasized because none of these features can be explained with the one-center scheme.

3.2. Modification of the beam profile in terms of the light intensity

Simulations of the photorefractive beam-distortion (along X -axis) during Y -axis propagation have been carried out using Crank-Nicolson finite-differences Beam Propagation Method [31,32]. In brief, the waveguide is considered as a set of very thin strips perpendicular to the propagation direction Y . Starting from the X -axis profile of the incident beam, the photorefractively modified refractive index profile of the first strip is determined using expression (4) for $|\Delta n_{\text{sat}}(I)|$. Then, the beam is Y -propagated through the strip using the calculated index profile. The procedure is successively repeated for the next strips along the propagation length. As propagation and refractive index change processes are mutually dependent processes the corresponding calculations have been iterated in each strip if necessary until consistency.

In order to compare with the BPM results, transversal beam profiles have been experimentally measured after the gaussian beam (waist diameter $90 \pm 10 \mu\text{m}$) has propagated a length of $3 \pm 1 \text{ mm}$ along a z-cut α -phase proton-exchanged LiNbO₃ waveguide fabricated on a congruent substrate. The guide has a gaussian refractive index profile, a surface index change of 10^{-2} ($\lambda = 532 \text{ nm}$) and an effective thickness of $2 \mu\text{m}$. The beam is coupled into the guide using a rutile prism-coupler and out-coupled with a second prism. The output beam propagates 80 mm in air and its steady state profile (which is reached in times ranging between 1 and 100 s, depending on the corresponding intensity,) is finally measured with a beam-profiler. Simulations have been performed with this experimental configuration using the same material parameters as in previous calculations and $T = 305 \text{ K}$. In Fig. 3, the measured (sequence (a)) and simulated (sequence (b)) transversal beam profiles have been represented as XZ density-plots for the light (mean) intensities indicated in the figure, namely 20 W/cm^2 , 250 W/cm^2 , 690 W/cm^2 and 1340 W/cm^2 . In sequence (c), the beam profiles (measured and simulated) have been represented graphically along X-coordinate. For $I = 20 \text{ W/cm}^2$ there is no appreciable beam-distortion because it is below (although close to) the so-called optical damage threshold [12]. For $I = 250 \text{ W/cm}^2$ some broadening is already observed due to the light-induced photorefractive index profile. At higher powers, a profile break-up appears due to the phase and intensity profiles imprinted in the beam by the photorefractive index change. Even more peaks have been observed at higher intensities, although they are somewhat worse defined than in Fig. 3.

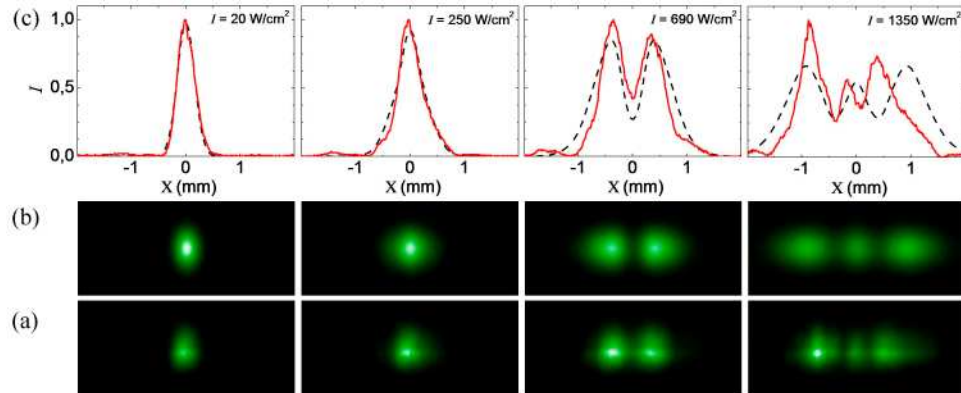


Fig. 3. Measured (a) and calculated (b) density plots of the transversal section of the beam obtained at the screen. c) Simulated (dashed lines) and measured (continuous lines) beam profiles corresponding to (a) and (b) spots respectively. The intensity of the incident beam inside the waveguide is indicated in the figure. The beam propagates 3 mm inside the guide, 3 mm through the prism coupler and 80 mm in the air. The configuration and the incident intensity are the same for the simulated and the experimental pictures.

It is remarkable the good accordance between theory and experiment that relies in the particular dependence $\Delta n(I)$ of the two-center photorefractive model. In fact, it has been further checked that the profile break-up is not merely produced by an increase of $|\Delta n_{\text{sat}}|$. To this end, beam propagation has been simulated for the same experimental configuration in an “idealized” sample with zero concentration for the secondary centre (Nb_{Li}), but a Fe concentration of $2 \times 10^{25} \text{ m}^{-3}$. Under these conditions, the plateau of Fig. 1 reaches a value of $|\Delta n_{\text{sat}}| \approx 10^{-3}$, even higher to that of $I = 1350 \text{ W/cm}^2$ in Fig. 3. However, non profile break-up was observed, emphasizing the essential role of the nonlinearity of the two-centre scheme for the appearance of a profile structure.

The simulations show that the general trend of the output profiles as the beam intensity increases is to reproduce the same sequential features: unaffected beam-profile, self-defocusing, profile break-up into two peaks and progressive increase of profile peaks. This

behavior is the same when the values of the simulation parameters are moved around. However, profile details such as the peak width and contrast or absolute intensity for a particular shape) are very sensitive to material parameters of the second center and geometrical details (input beam width, propagation length ...).

4. Discussion

The simulations and experimental measurements presented above represent a successful application of the two-center photorefractive model (Fe and Nb_{Li} defects) to the single beam configuration. A good agreement has been found between simulated and experimental data on the beam-intensity dependence of three central aspects: induced refractive index change Δn_{sat} , photovoltaic current j_{pv} and transversal beam profile. For the last one, a beam propagating method which includes the model equations has been used. In addition, the full curve conductivity-versus-intensity has been simulated and favorably compared with published experimental data, although only qualitatively because of the lack of enough experimental details. From all these data, a refined set for the material parameters, which improves the agreement between simulated and measured data, is proposed.

As already stated above, the key point in the optical damage response is the effect of the secondary shallow-center. The photovoltaic efficiency of the Nb_{Li}⁴⁺-state is about five times greater than that of the Fe²⁺. However, its important influence only is apparent at high enough beam-intensities and/or low temperatures, since otherwise only the Nb_{Li}⁵⁺ is present. This naturally explains the increase of the refractive index change with light intensity which is the origin of beam self defocusing. At low intensities, thermal electron detrapping from Nb_{Li}⁴⁺ overwhelming dominates over electron-Nb_{Li}⁵⁺ recombination and no detectable contribution to the photocurrent is observed. Only at relatively high intensity with a high rate of Fe²⁺-excitation, or low temperatures with a low thermal excitation rate, the population increase of Nb_{Li}⁴⁺ makes relevant its photovoltaic contribution and its saturating effect on σ . The essential role of the strong nonlinearity induced by the Nb_{Li}-defect [20,23] has also been proved, as no beam-profile structure appears in steady state when this defect is absent, even with a high Fe concentration. Under this picture, the methods for inhibiting the wavefront-distortion can be easily understood. It should decrease when the anti-site concentration diminishes (stoichiometric or Mg-doped crystals) [1,7] and/or when, due to sample heating [12], the Nb_{Li}⁴⁺-population decreases.

These results joined to those previously reported in [18], provide a comprehensive description of photorefractive optical damage. There, the two-center model was successfully applied to the recording of holographic gratings in order to understand holographic induced nonlinear scattering. The strong nonlinearity of the grating amplification-gain was correlated with the appearance of intensity-thresholds for catastrophic optical damage obtaining good accordance with experimental data in a semi-quantitative way. Other features, here confirmed with the single-beam configuration, were found there: the effect of the Nb_{Li}-concentration on the intensity dependences of Δn (to increase diffraction efficiencies) and the role of the temperature.

To further determine the usefulness of the model in common cases and go into the underlying physics, let us write below the limitations of the model:

- a. It is a two-dimensional approach strictly appropriate for planar waveguides although the main features should also apply to bulk crystals
- b. At very high intensities, the free carrier concentration is no longer negligible (as assumed in the model) with regard to the donor or acceptor concentrations.
- c. Extreme-oxidation states, as those reported in [13] for reducing the optical damage, cannot be treated with the present state of the model.

d. It is pending the application of BPM to the case mentioned above of holographic amplification of optical noise.

In conclusion, despite the above limitations of the model, this paper provides a convincing and solid support for considering the two-center model of general applicability in most common optical damage problems. In addition, since it is based on sound microscopic basis, it constitutes a powerful and reliable tool for handling the problem of optical damage when designing photonic devices.

Acknowledgements

This work was supported by the Ministerio de Educación y Ciencia (MEC) under grant MAT2008-06794-C03. J. Villarroel acknowledges his FPI fellowship from MEC. We also thank Dr. O. Caballero-Calero for useful discussions and comments.